Breast Cancer Prediction with XAI (Explainable AI) and Risk Score Using Streamlit Dashboard

This project is about predicting whether a breast tumor is benign (not cancerous) or malignant (cancerous) using machine learning techniques. The goal is not only to make accurate predictions but also to help doctors, healthcare workers, and even patients understand why the model made a certain prediction.

Usually, machine learning models work like a black box – they give an answer but don't explain how they arrived at that answer. This can be a problem, especially in healthcare where decisions need to be trusted and understood clearly.

To solve this, I have used an Explainable AI technique called **SHAP (SHapley Additive exPlanations)**. SHAP helps explain the predictions by showing the impact of each feature (like age, tumor size, etc.) on the prediction. This way, users can see both:

- **Global explanations** – which features are most important for predictions overall.
- **Local explanations** – why the model made a specific prediction for one patient.

Additionally, I created an interactive "**Streamlit dashboard**"where users can enter their own data, test the prediction, see the risk score, and get explanations.

Project Workflow:

1. **Data Loading:** The dataset provided by the company was loaded into Google Colab.

2. **EDA:** Performed Exploratory Data Analysis with visualizations like correlation heatmaps and feature distributions.

3. **Model Training:** Compared multiple models (Logistic Regression, Decision Tree, Random Forest, SVM, KNN, XGBoost).

4. **Model Selection:** Chose the best model based on accuracy and AUC scores.

5. **Calibration:** Applied isotonic calibration to convert probabilities into calibrated risk scores (0–100).

6**. Explainability:** Used SHAP to explain feature importance globally and locally.

7. **Streamlit App:** Built an interactive dashboard for predictions, risk scores, and SHAP explanations

Step-wise Results and Outputs:

Step 1 – Loading and Exploring the Dataset

Output:

- The dataset has 569 rows and 32 columns.
- The columns include patient ID, diagnosis, and various tumor features like radius_mean, texture_mean, area_mean, etc.

Dataset Preview:

The first few rows show sample data where the diagnosis is either 'M' (malignant) or 'B' (benign), along with feature values such as radius_mean = 17.99, area_mean = 1001.0, etc.

Dataset Overview:

-A table showing initial rows and columns.

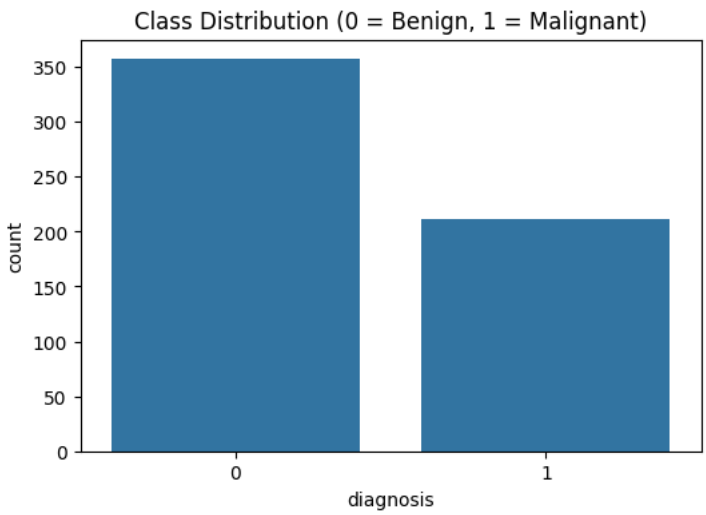
```
Choose Files data.csv
data.csv(text/csv) - 125141 bytes, last modified: 8/23/2025 - 100% done
Saving data.csv to data.csv
Shape: (569, 32)
Columns: ['id', 'diagnosis', 'radius_mean', 'texture_mean', 'perimeter_mean', 'area_mean', 'smoothness_mean', 'compactness_mean', 'concavity_mean', 'concave points_mean', 'symmetry_mean', 'fractal_dimension_mean', 'radius_se',
        id diagnosis  radius_mean  texture_mean  perimeter_mean  area_mean  \
0   842302        M        17.99         10.38          122.80     1001.0
1   842517        M        20.57         17.77          132.90     1326.0
2  84300903        M        19.69         21.25          130.00     1203.0
3  84348301        M        11.42         20.38           77.58      386.1
4  84358402        M        20.29         14.34          135.10     1297.0

   smoothness_mean  compactness_mean  concavity_mean  concave points_mean  \
0          0.11840           0.27760          0.3001              0.14710
1          0.08474           0.07864          0.0869              0.07017
2          0.10960           0.15990          0.1974              0.12790
3          0.14250           0.28390          0.2414              0.10520
4          0.10030           0.13280          0.1980              0.10430

   ...  radius_worst  texture_worst  perimeter_worst  area_worst  \
0  ...         25.38          17.33           184.60      2019.0
1  ...         24.99          23.41           158.80      1956.0
2  ...         23.57          25.53           152.50      1709.0
3  ...         14.91          26.50            98.87       567.7
4  ...         22.54          16.67           152.20      1575.0
```

-The class distribution:
diagnosis 0 (benign): 357
diagnosis 1 (malignant): 212.

```
Class distribution:
diagnosis
0    357
1    212
Name: count, dtype: int64
```



Step 2 – Data Cleaning and Preparation

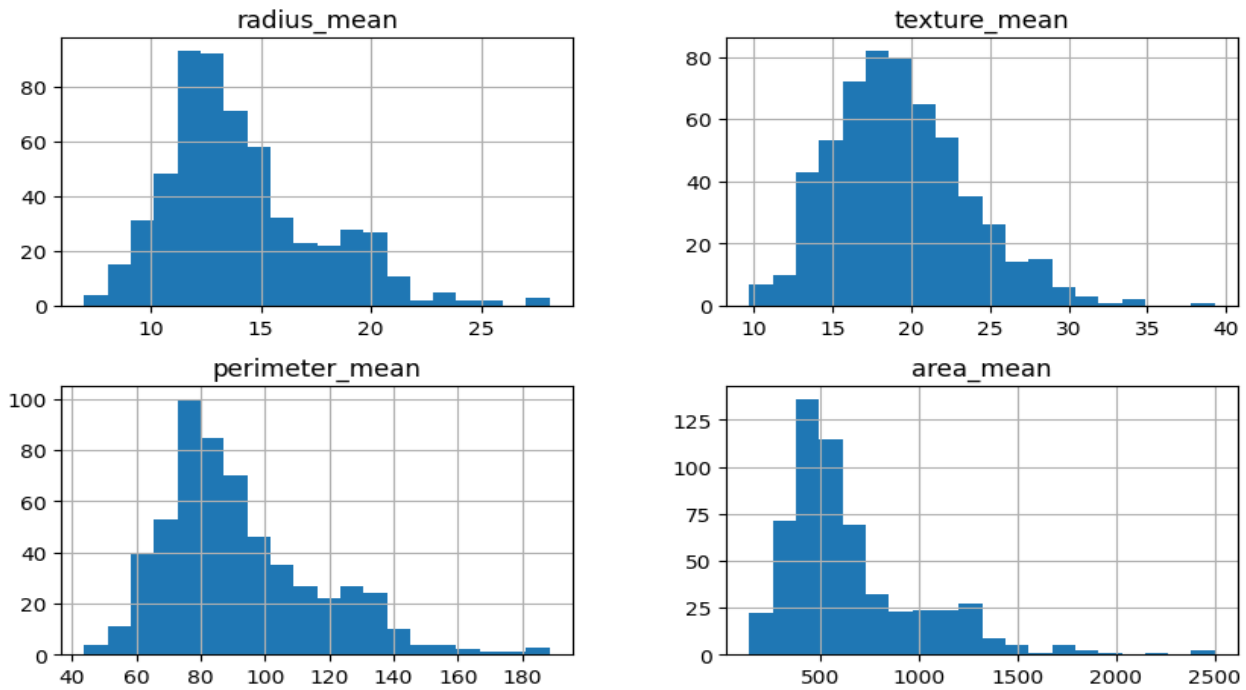
1. Removed unnecessary columns like id.
2. Converted diagnosis from letters ('M', 'B') to numeric values (1 = malignant, 0 = benign).
3. Checked for missing values and normalized the data to bring all features to a similar scale.

Output:

- Data ready for machine learning.
- Class balance verified to ensure no bias.

Data Cleaning Preview:

-Cleaned dataset with columns like radius_mean, texture_mean, etc.



-Processed data fed into training and testing sets.

Step 3 – Splitting Data into Train and Test Sets

Output:

- 1.Training set: 455 samples
- 2.Test set: 114 samples

```
Train shape: (455, 31)
Test shape: (114, 31)
```

Step 4 – Model Training and Accuracy Comparison

Models Used:

A. Logistic Regression
B. Decision Tree
C. Random Forest
D. Support Vector Machine (SVM)
E. K-Nearest Neighbors (KNN)
F. XGBoost

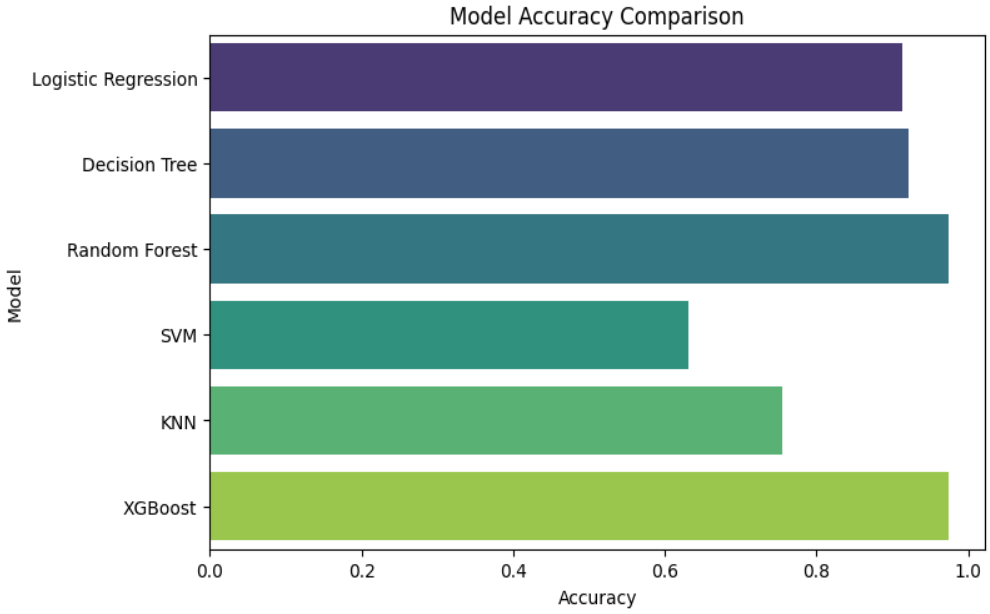
Results:

Best Model: XGBoost, with an accuracy of 97.37%.

```
Comparison Table:
                 Model  Accuracy
0  Logistic Regression  0.912281
1        Decision Tree  0.921053
2        Random Forest  0.973684
3                  SVM  0.631579
4                  KNN  0.754386
5              XGBoost  0.973684
```

-A bar graph comparing the accuracy of each model.

-XGBoost and Random Forest performing the best.




```
Best Model: XGBoost
```

Step 5 – Model Evaluation

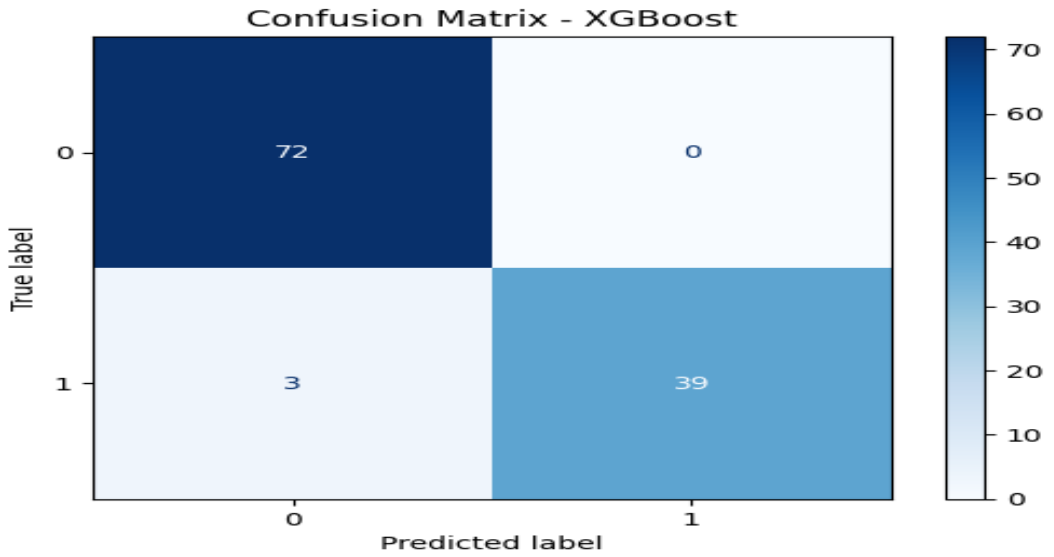
This step is about checking how well the model performs using standard metrics. These help us understand whether the predictions are accurate, how well the model distinguishes between classes, and whether the probability outputs are reliable.

Step 5.1 – <u>Confusion Matrix</u>

What is it?
The confusion matrix helps us see how many predictions are correct and how many are wrong. It tells us:

1. How many malignant tumors were predicted correctly.
2. How many benign tumors were predicted correctly.
3. The number of false alarms and missed cases.

Output :
A table or heatmap showing values like True Positives, True Negatives, False Positives, and False Negatives.

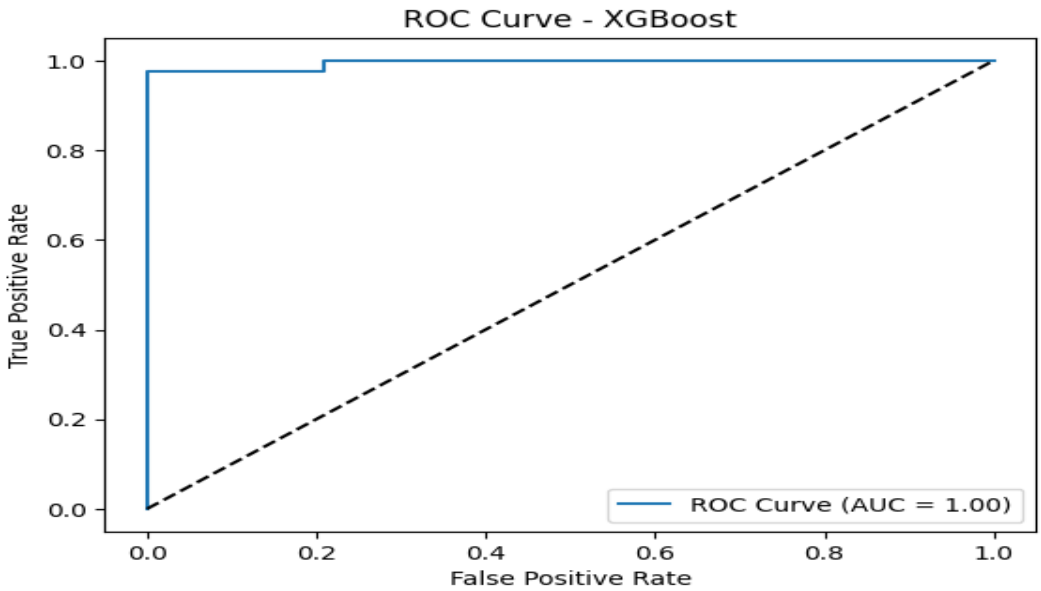


Step 5.2 – <u>ROC Curve</u>

What is it?
The ROC curve shows how well the model distinguishes between benign and malignant tumors at different thresholds.

1. The closer the curve is to the top-left corner, the better the model's performance.
2. The area under the curve (AUC) shows the overall effectiveness of the model.

Output:
A graph plotting True Positive Rate vs. False Positive Rate with the curve and AUC score displayed.

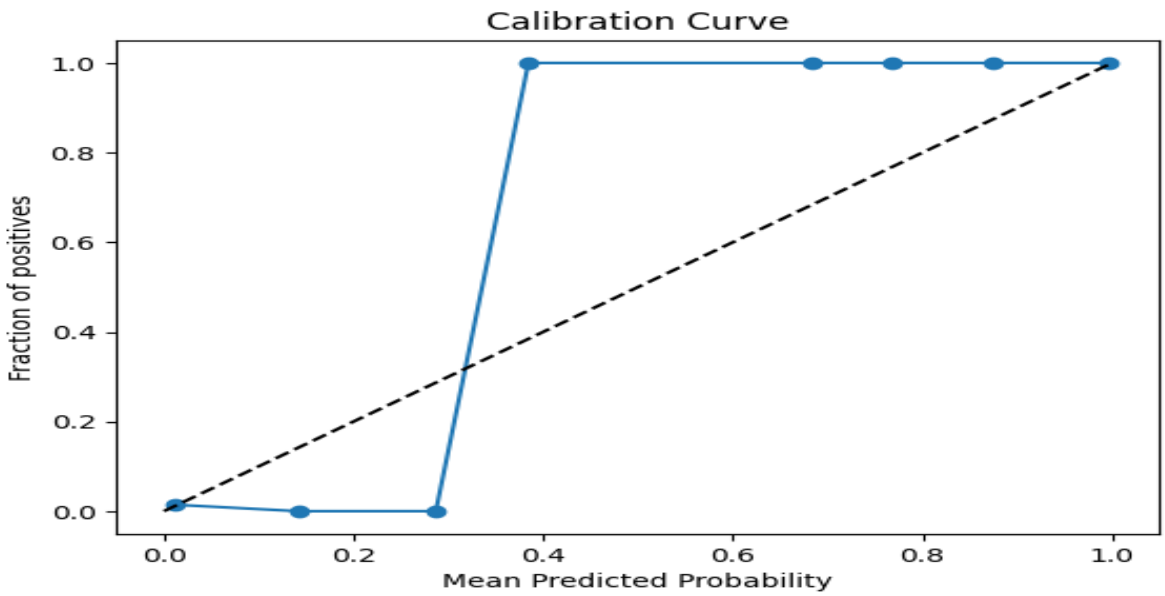


Step 5.3 – <u>Calibration Curve</u>

What is it?
The calibration curve compares predicted probabilities with actual outcomes. It helps us understand if the predicted risks (like 80% or 20%) are meaningful.A curve close to the diagonal line means predictions are well-calibrated.

Output:
A plot comparing predicted probabilities with actual outcomes.

```
Sample Calibrated Risk Scores (0-100):
[  0.95 100.    13.15 100.     5.64   0.95  91.94   0.95   0.15   0.95]
```



Step 5.4 – <u>Calibrated Risk Scores</u>

What is it?
The calibrated risk score shows how much risk the model assigns to each patient. It's easier to interpret than raw probability outputs.

a) For example, a risk score of <u>0.95%</u> means very low risk.

b) A score of <u>100%</u> means very high risk.

```
Calibrated risk for sample index 0: 0.95 %
Predicted class (threshold 0.5): Benign
```

Step 6 – Explainable AI Using SHAP

This step explains why the model made its predictions. It helps doctors, researchers, and users understand which features contributed to a diagnosis.

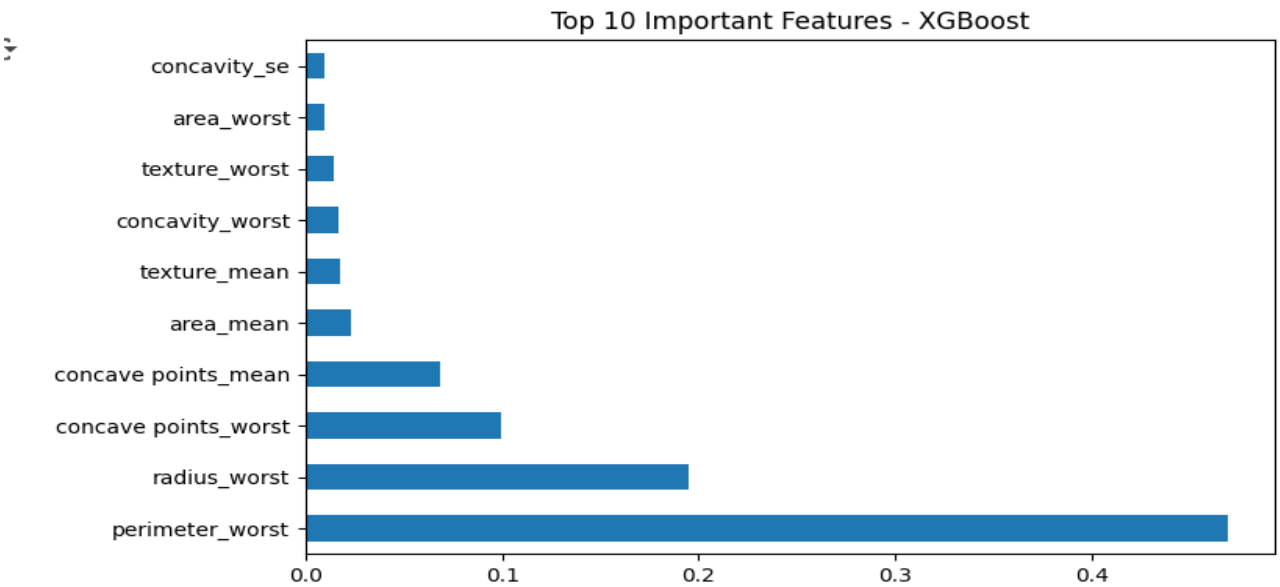
Step 6.1 – <u>Global SHAP Explanation</u>

What is it?
The global SHAP explanation shows which features influence predictions across the entire dataset.

A. Features like radius_mean, area_mean, etc., are ranked by their importance.

B. It helps users know which medical measurements matter most.

Output
A summary plot or feature importance chart highlighting the top features.



Note:
There was an error while creating the local SHAP plot because the model's prediction was a single value instead of an array of SHAP values. The error message shown was:

```
Plotting SHAP summary (global importance)...
Could not create SHAP summary plot: Summary plots need a matrix of shap_values, not a vector.
```

<u>Step 6.2 – Local SHAP Explanation</u>

What is it?
The local SHAP explanation shows how each feature affected the prediction for one patient.

A. It visualizes which measurements pushed the prediction toward benign or malignant.
B. For example, a high radius_mean may increase the risk score.

Note:
In this project, the local SHAP explanation could not be plotted because the model's output was a single value (a float) instead of a list or array that SHAP expected. As a result, this error appeared:

```
Local SHAP for selected sample (waterfall):

Local SHAP failed: object of type 'numpy.float32' has no len()
```

Step 6.3 – <u>How SHAP Helps in Understanding the Model's Predictions</u>

SHAP helps explain how the machine learning model makes its predictions by showing the contribution of each feature. For example, it shows whether a high radius_mean or a low texture_mean increased or decreased the risk of cancer. This makes it easier for doctors and healthcare workers to understand and trust the results, especially when making important decisions about a patient's treatment.

Step 7 – Streamlit Dashboard

This part of the project gives a user-friendly way to interact with the model. Instead of just running code, users can input values and see predictions, explanations, and risk scores instantly. Below is a breakdown of each part of the dashboard.

The app shows:

1. **Prediction (Benign or Malignant)-** based on a threshold (0.5).

2. **Calibrated Risk Score-** between 0 and 100.
3. **SHAP explanations** (though some may fail due to data formatting).

➢ **Input Features Section**

On the left side of the dashboard, there are several boxes where users can enter or adjust values related to a tumor's characteristics. These are called features and help the model predict whether the tumor is benign or malignant.Here are some of the features explained in simple terms:

1. **id**
   Every patient or record has a unique ID number. This helps keep track of data but doesn't affect the prediction.
2. **radius_mean**
   This is the average size of the tumor's radius (distance from the center to the edge). Larger values usually suggest more concern.
3. **texture_mean**
   It shows how rough or smooth the tumor surface is on average. Irregular textures can indicate malignancy.
4. **perimeter_mean**
   The average boundary length of the tumor. Bigger tumors usually have a longer perimeter.
5. **area_mean**
   The average area of the tumor in square units. Bigger tumors are more likely to be dangerous.
6. **smoothness_mean**
   How smooth the edges of the tumor are. Tumors with uneven edges can be more suspicious.
7. **compactness_mean**
   This shows how dense or spread out the tumor is. More compact tumors are easier to classify.
8. **concavity_mean**
   It measures how much the tumor's shape curves inward. Irregular shapes are often linked with cancer.
9. **concave points_mean**
   The number of inward curves or points on the tumor's edge. More concave points suggest abnormal growth.

There are other features too, but these are the most important ones that users can adjust for testing predictions.

➢ **Prediction Section**

Once the user enters the values, the dashboard shows:

1. **Predicted Class – Benign or Malignant**
   The model uses the inputs and predicts whether the tumor is harmless (Benign) or harmful (Malignant).
   In the example screenshot, it shows **Malignant**, meaning the model thinks the tumor is likely cancerous.
2. **Calibrated Risk Score (0–100)**
   This number gives a clearer understanding of how risky the tumor is.
   In the example, it shows **100**, meaning the model is highly confident the tumor is malignant. A score close to 0 would mean very low risk.This score helps doctors know how seriously they should take the results.

➢ **SHAP Feature Importance (Local Explanation)**

This section explains **why** the model made the prediction. It uses a method called **SHAP (SHapley Additive exPlanations)** that tells how each feature contributed.

What it means:

- Positive values → Push the prediction towards **Malignant**.
- Negative values → Push the prediction towards **Benign**.

This way, doctors can see which features increased or decreased the risk.

➢ **Graph – SHAP Waterfall Plot**

The waterfall chart shows how each feature contributed to the final prediction.In simple terms:

✓ Each bar represents one feature.

✓ The longer the bar, the more influence it had on the prediction.

✓ Red bars push the result towards malignancy.

✓ Blue bars push it towards benign.

For example, if radius_mean is very high, it may push the prediction strongly towards malignant.

In the example screenshot, even though some of the values are zero, the graph visually explains how the decision was made based on tumor features.

- **What is "f(x)" in the SHAP Graph?**

In the waterfall plot shown in the dashboard, you can see something like this at the top of the graph:f(x)=1

This means:

1. **f(x)** is the final output of the machine learning model for this particular patient.
2. The value **1** here stands for the **Malignant** class (in this project, 1 = Malignant, 0 = Benign).

- **What does this tell us?**

1. The model looks at all the features (like radius_mean, texture_mean, etc.) and combines them to give one final result.
2. The graph shows how each feature either increased or decreased the chance of the tumor being malignant.
3. So, **f(x) = 1** means that after considering all features, the model predicts that the tumor is malignant.

- **Why it's useful?**

Doctors can not only see the final result but also understand **why** the model thinks the tumor is cancerous. The graph visually explains how the numbers for each feature contributed to reaching this conclusion.

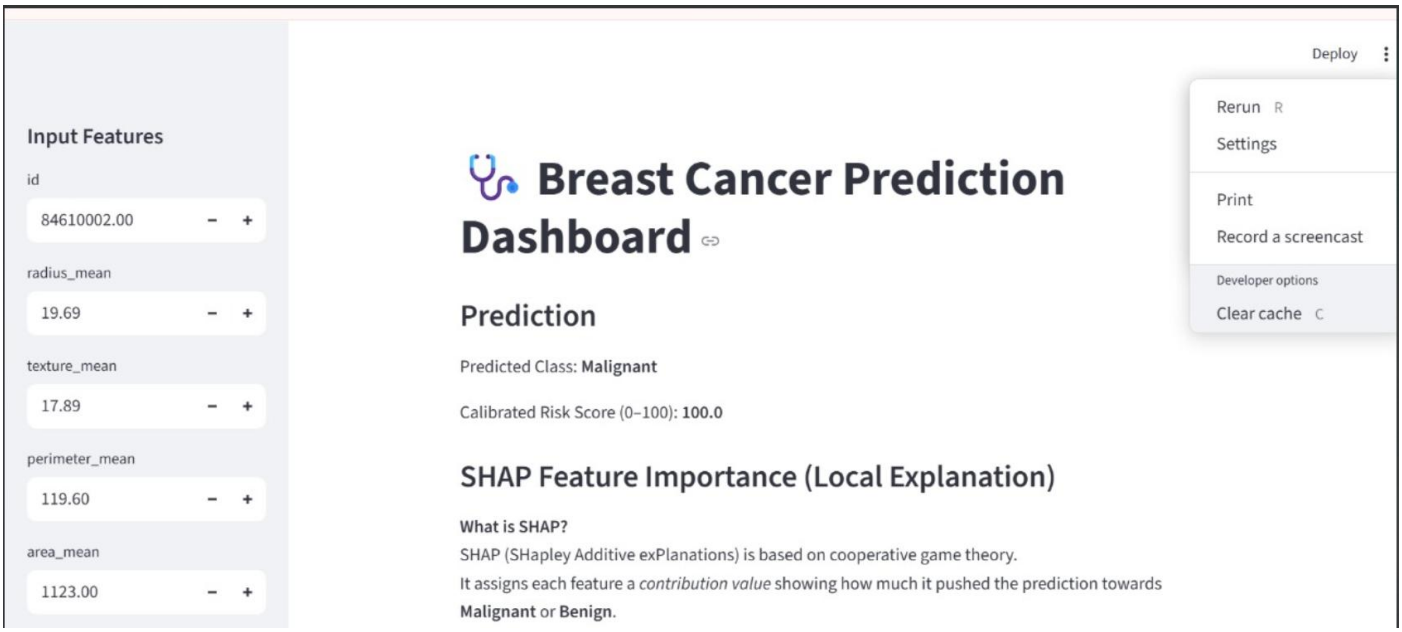
- **Explanation of "f(x)" in the SHAP Graph:**

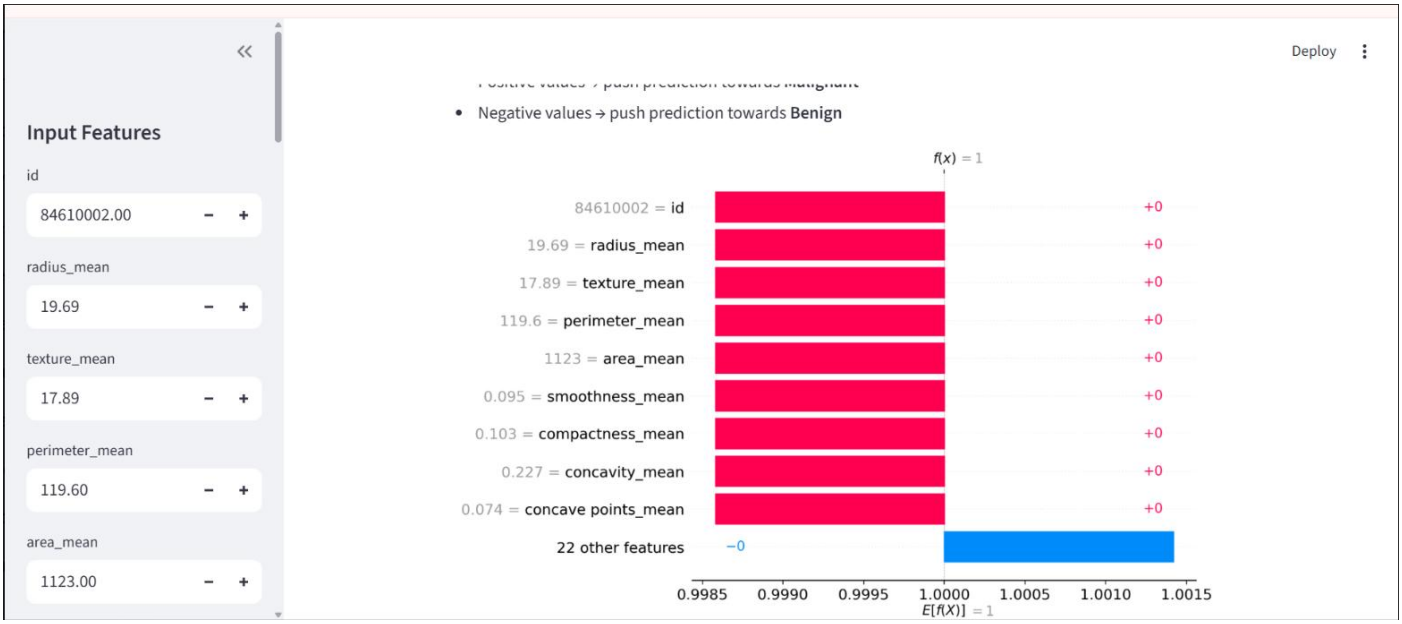
In the graph, **f(x) = 1** means that the model's final prediction for this patient is **Malignant**. The model takes all the input features and adds their effects together to give this result. The graph helps us see how each feature pushed the prediction towards malignant or benign, making the process easier to understand.

- **Why This Dashboard is Helpful**

1. Doctors can input patient details and get predictions instantly.
2. Non-technical users can see how each feature affects the result instead of being confused by just a number.
3. Patients can better understand their condition.
4. The risk score helps in making decisions confidently.

It’s a simple yet powerful way to apply machine learning in healthcare while keeping the process transparent.





Final Summary:

1. The project successfully predicts breast cancer using machine learning models, with XGBoost giving the best results.
2. SHAP was applied to explain predictions globally and locally, although some plots failed due to technical issues.
3. The calibrated risk score makes the prediction easier to interpret.
4. The Streamlit dashboard provides an interactive way to test and visualize predictions, helping non-technical users understand the model’s output.